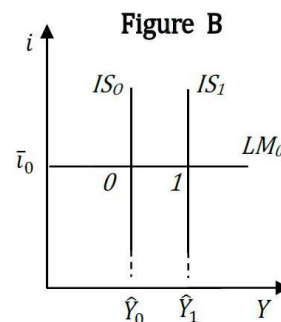
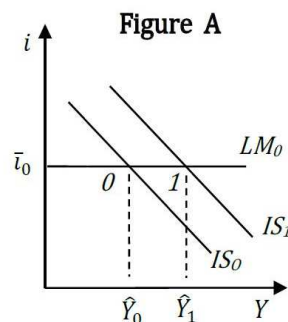


Question 1 (7.5 points)

In country Macro, described by the IS-LM model, government purchases of goods and services, net taxes and investment are exogenous ($G = \bar{G}$, $T = \bar{T}$ and $I = \bar{I}$), while the consumption function is $C = c_0 + c_1(Y - \bar{T}) - h_2 i$, where the parameter $h_2 > 0$ is the sensitivity of consumption to the interest rate. Finally, the central bank chooses the interest rates. When Macro was in an initial equilibrium denoted by '0' in the figure, the government of the country cuts net taxes and the economy reaches the new equilibrium '1'. Given the above assumptions, which of the two following figures, A or B, represents the economy of Macro and the new equilibrium it will reach after the fiscal policy experiment just described? Explain the reasons for the changes in equilibrium income, consumption and investment that will take place in the move from '0' to '1'. As the economy goes from the first to the second equilibrium, how must the changes in consumption and in income be related to one another?



SKETCH OF ANSWER

Even though investment does not depend on it, aggregate demand is nevertheless decreasing in the interest rate, since an increase in this latter variable lowers consumption. The IS curve has therefore, as usual, a negative slope (if, starting from a goods market equilibrium position, i falls, in this economy C will rise, leading to an excess demand for goods; to return to equilibrium, the supply of goods, Y , will have to go up), and Figure A is the correct representation of Macro's economy. If T decreases, the IS curve shifts to the right, since the increase in disposable income caused by the tax cut raises consumption, and therefore aggregate demand, for any given level of Y . The equilibrium point becomes 1, with a higher level of output. Being G and I (in this economy, both exogenous) unchanged, consumption must have gone up. Since in this new equilibrium supply and demand must be equal, and consumption is the only component of aggregate demand that has changed, in the move from '0' to '1' C must have risen by the same amount by which Y , the supply of goods, has gone up.

Question 2 (7.5 points)

Suppose now that Macro's central bank intends to bring back equilibrium income to Y_0 , the level prevailing before the tax cut described in the previous question. In your opinion, choosing a new level for the interest rate would allow it to achieve its goal? If so, compare the composition of aggregate demand in the new equilibrium that would be reached following the central bank's monetary policy intervention with that prevailing in the initial equilibrium '0'. If, instead, you think that changing the interest rate would not bring back income to Y_0 , carefully explain why.

SKETCH OF ANSWER

The central bank will have to raise the interest rate. Since, in the new equilibrium, income is back at the value it was taking on at '0', the same must be true about aggregate demand. Being both G and I exogenous, this requires consumption to be, in the final equilibrium, the same as in the initial one, point '0' – the favorable impact of the decrease in net taxes, which tends to raise C , must have been exactly offset by the adverse impact that, in this economy, a higher interest rate has on consumption.

Question 3 (7.5 points)

True or False?

Explain whether the following statement is true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

“The *zero lower bound* for the nominal interest rate implies that, if expected inflation is -3% , then the real interest rate cannot be smaller than -3% ”.

SKETCH OF ANSWER

False. The real interest rate (r) is the difference between the nominal rate (i) and expected inflation. Given expected inflation, r is lowest when $i = 0$, so that r cannot fall below minus the expected rate of inflation. It follows that, being expected inflation -3% in the present setting, the real interest rate cannot be less than 3% .

Question 4 (7.5 points)

By making reference to the graph “supply of goods-demand for goods” used to analyse the determination of the medium-run equilibrium, natural real interest rate, r_n , explain if and how r_n will change if, following a change in preferences, autonomous consumption c_0 goes up. In addition, compare the composition of aggregate demand prevailing in the new medium run equilibrium that will be reached after this change with that prevailing before the increase in c_0 . Explain.

SKETCH OF ANSWER

An increase in autonomous consumption leads to an increase in the amount of goods demanded for any given levels of income and the real interest rate. In the graph "supply of goods-demand for goods" the demand curve shifts to the right, while the supply curve remains unchanged. It follows that, in the new medium run equilibrium, the real interest rate will go up by the amount needed to make sure that the demand for goods remains unchanged, just like the supply of goods. Finally, in this new equilibrium consumption will be higher (disposable income is unchanged, but higher), investment lower (by the same amount by which consumption has gone up) due to the increase in the real interest rate, and G , being exogenous, unchanged.